

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	K.Prathiba	Reg. No.:	192421163
Date:	23-08-2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

1. IDENTIFICATION AND SELECTION OF RELEVANT RESEARCH PAPERS

The proposed **AI-Based Personal Finance and Budget Planning System** focuses on managing income, expenses, budgets, savings, spending patterns, and personalized financial recommendations using Artificial Intelligence and Machine Learning. Therefore, research papers related to AI-based personal finance, expense tracking, predictive analytics, budget forecasting, and personalized financial recommendations were selected.

The selected papers were taken from research journals, conferences, scholarly publications, and academic research sources. The papers were chosen based on their relevance to the proposed problem statement and their use of technologies such as Machine Learning, Deep Learning, Natural Language Processing, predictive analytics, and Explainable AI.

2. SUMMARY OF OBJECTIVES, METHODOLOGIES, TECHNOLOGIES, DATASETS AND FINDINGS

2.1 Agarwal et al. (2024)

Research Problem

Traditional personal finance applications mainly provide basic expense tracking and budgeting. They provide limited personalized financial guidance.

Objective

The objective was to develop an **AI-powered personal finance assistant** that could help users manage finances and improve financial literacy.

Methodology

The system combines:

- Artificial Intelligence
- Machine Learning
- Natural Language Processing
- Expense analysis
- Personalized recommendations

The system analyses user financial information and generates financial insights.

Technologies

The research uses technologies including:

- React.js
- Flask

- MongoDB
- Firebase
- Machine Learning
- Natural Language Processing

Dataset

The system works with user-provided financial information such as income, expenses and financial activities.

Major Findings

The research demonstrates that AI can provide more personalized financial assistance than conventional finance tracking systems.

Relevance

This paper is directly related to the proposed system because the proposed project also aims to provide personalized financial recommendations.

2.2 Kumar et al. (2024)

Research Problem

Users often have difficulty predicting future expenses using conventional expense management applications.

Objective

The research aims to develop a personal finance manager that uses **predictive analytics** to support financial decision-making.

Methodology

The study uses **Long Short-Term Memory (LSTM)** networks.

LSTM is useful for analysing sequential data because financial transactions occur over time.

The system uses historical financial records to identify spending trends and predict future financial behaviour.

Technologies

- Machine Learning
- LSTM
- Predictive Analytics
- Data Visualization

Dataset

Historical income and expense information is used for financial prediction.

Major Findings

The study shows that historical financial data can be used to predict future spending and support better financial decisions.

Limitation

Prediction performance depends heavily on the availability and quality of historical data.

Relevance

The proposed project can use predictive analysis to identify future spending patterns and recommend suitable budgets.

2.3 Bhavana et al. (2025)

Research Problem

Traditional financial systems are often reactive. They show what users have already spent instead of proactively helping them plan future finances.

Objective

The research proposes an **AI-based wealth advisory system** for:

- Personalized budget planning
- Financial goal setting
- Expenditure optimization
- Anomaly detection
- Personalized recommendations

Methodology

The system uses:

- Classification
- Forecasting
- Anomaly detection
- Explainable AI
- Natural Language Generation

Dataset

Financial transaction and user financial information are used for analysis.

Major Findings

The study reports improved financial planning through AI-based personalized recommendations. It also reports high anomaly-detection performance in its pilot evaluation.

Limitation

The research requires wider real-world testing with larger and more diverse users.

Relevance

The research supports the proposed project's objective of providing personalized budget recommendations.

2.4 Amrendra (2025)

Research Problem

Users frequently perform many digital transactions, making manual expense tracking difficult.

Objective

The **Smart Pocket** system aims to automate:

- Expense tracking
- Expense classification
- Spending prediction
- Financial visualization

Methodology

Machine Learning techniques are used to classify expenses and predict spending patterns.

Technologies

- Machine Learning
- Classification
- Predictive Analytics
- Data Visualization

Dataset

Expense and transaction records are used.

Major Findings

The research shows that Machine Learning can automatically organize expenses and help users understand their spending patterns.

Limitation

The effectiveness of prediction depends on the quality and quantity of historical transaction data.

Relevance

The proposed project can extend this approach by adding personalized budget planning and AI recommendations.

2.5 Zothansياما et al. (2025)

Research Problem

Traditional expense trackers record financial transactions but generally do not provide intelligent budgeting recommendations.

Objective

The objective is to develop a **Personal Finance Tracker with Budget Recommendation AI**.

Methodology

The research uses:

- Heuristic scoring
- Trend analysis
- Predictive techniques
- Expense analysis

Dataset

User financial transactions and expense information are used.

Major Findings

The system provides:

- Budget recommendations
- Saving suggestions
- Spending analysis
- Financial reports

Limitation

Lightweight recommendation methods may not adapt sufficiently to complex or rapidly changing financial behaviour.

Relevance

This is one of the closest studies to the proposed project because both focus on expense tracking and AI-based budget recommendations.

2.6 Garugu et al. (2026)

Research Problem

Manual financial data processing is time-consuming and makes it difficult for users to understand future financial requirements.

Objective

The study proposes an intelligent personal finance management system that provides:

- Automated financial data processing
- Expense categorization
- Budget forecasting
- Financial analysis
- Predictive insights

Methodology

The research combines:

- Machine Learning
- Regression
- Time-series forecasting
- Automated data processing

Dataset

Transaction and financial records are analysed.

Major Findings

The study shows that automated financial processing combined with predictive analytics can improve personal financial planning.

Limitation

Prediction accuracy can be affected by incomplete or inconsistent financial data.

Relevance

The proposed system can use similar techniques for automated budget planning.

2.7 Saranya et al. (2026)

Research Problem

Manual recording of financial transactions requires considerable user effort.

Objective

The objective is to develop an AI-powered expense management system capable of automatically processing financial information.

Methodology

The research combines:

- Artificial Intelligence
- Machine Learning
- Natural Language Processing
- Optical Character Recognition
- Predictive Analytics

Dataset

The system can process transaction and receipt-related financial information.

Major Findings

The study demonstrates the possibility of automating expense tracking, categorization and spending prediction.

Limitation

OCR-based expense extraction can be affected by:

- Poor-quality images
- Different receipt formats
- Missing information
- Handwritten content

Relevance

OCR can be considered as a future enhancement for the proposed system.

2.8 Chahar et al. (2026)

Research Problem

Many existing personal finance systems use fixed budgeting methods and generic recommendations.

Objective

The objective is to develop an AI-powered personal finance system that provides dynamic and personalized financial insights.

Methodology

The system combines:

- Machine Learning
- Natural Language Processing
- Personalized recommendations
- User-context analysis

Dataset

Financial and user-context data are used for analysis.

Major Findings

The research indicates that AI can provide more personalized financial insights than static budgeting systems.

Limitation

More real-world and long-term evaluation is required.

Relevance

The paper supports the proposed system's goal of providing dynamic financial recommendations.

3. COMPARISON OF EXISTING APPROACHES

The existing research approaches can be compared based on:

- Methodology
- Prediction capability
- Personalization
- Accuracy
- Efficiency
- Scalability
- Explainability
- Data requirements

3.1 Methodology Comparison

Approach	Methodology	Main Purpose
Rule-Based	Predefined rules	Basic budgeting
Heuristic	Scoring and thresholds	Budget recommendation
Machine Learning	Classification/regression	Expense analysis
LSTM	Deep learning/time series	Spending prediction
NLP	Language processing	Financial assistance
OCR + ML	Receipt extraction + classification	Automated expense entry
XAI	Explainable prediction	Transparent recommendations

Analysis

Rule-based methods are simple and easy to understand, but they provide limited personalization.

Machine Learning provides better pattern recognition and can learn from historical financial information.

LSTM-based systems are suitable for time-series prediction but require more historical data.

NLP-based systems are useful for conversational financial assistance.

Explainable AI is especially useful in financial applications because users need to understand the reasons behind recommendations.

3.2 Accuracy Comparison

Accuracy is particularly important in:

- Expense classification.
- Spending prediction.
- Anomaly detection.
- Budget recommendations.

Machine-learning approaches can provide better accuracy than simple rules when sufficient training data is available.

However, accuracy varies according to:

- Dataset size.
- Data quality.
- Number of categories.
- User behaviour.
- Model selection.

Therefore, an AI model cannot guarantee the same accuracy for every user.

3.3 Performance Comparison

Simple rule-based systems generally have low computational requirements.

Machine-learning models require more computational resources but provide improved analytical capabilities.

LSTM and other deep-learning models require additional processing power and training time.

For a personal finance application, the proposed system should balance performance with prediction quality.

3.4 Scalability Comparison

A scalable system should support:

- Increasing numbers of users.
- Increasing transaction records.
- Multiple expense categories.
- Long-term financial histories.

Cloud databases and modular application architectures can support scalability.

The proposed system can use a modular design so that financial tracking, budget management and AI analysis can be developed independently.

3.5 Efficiency Comparison

Approach	Efficiency	Explanation
Rule-Based	High	Simple calculations
Heuristic	High	Low computational cost
ML	Medium–High	Requires model processing
LSTM	Medium	Higher computational requirement
NLP	Medium	Depends on language model
OCR + ML	Medium	Requires image processing
Hybrid AI	High potential	Combines appropriate methods

A hybrid approach can provide a good balance between computational efficiency and intelligent analysis.

3.6 Personalization Comparison

Traditional systems provide limited personalization.

AI-based systems can consider:

- User income.
- Expense history.
- Spending categories.
- Budget limits.
- Financial goals.

Therefore, AI provides better opportunities for personalized financial management.

4. LIMITATIONS AND CHALLENGES OF EXISTING APPROACHES

4.1 Dependence on Historical Data

Predictive models need sufficient historical transactions.

New users may not have enough data for reliable predictions.

Challenge

The system should still provide useful financial guidance when historical data is limited.

4.2 Data Quality

Incorrect or incomplete financial information can produce incorrect recommendations.

Examples:

- Missing transactions.
- Incorrect amounts.
- Wrong categories.
- Duplicate records.

Therefore, data validation is important.

4.3 Changing User Behaviour

A user's spending pattern may change suddenly.

For example:

- Job change.
- Emergency expenses.
- Travel.
- Education expenses.
- Seasonal purchases.

Historical models may fail to predict such changes.

4.4 Limited Explainability

Some advanced AI models behave like black boxes.

Users may receive a recommendation without understanding its reason.

This is a major issue in financial applications.

4.5 Privacy

Financial data is highly sensitive.

The system must protect:

- Income information.
- Expense information.
- Bank-related data.
- Personal details.
- Financial goals.

4.6 Security

Potential threats include:

- Unauthorized access.
- Account attacks.
- Data leakage.
- Weak authentication.
- Insecure APIs.

Security should therefore be integrated into system design.

4.7 AI Bias

AI recommendations may become biased if training data does not adequately represent different financial situations.

A recommendation suitable for a high-income user may not be suitable for a low-income user.

4.8 AI Hallucination

Generative AI may produce incorrect financial information.

Therefore, the proposed system should preferably generate recommendations from verified user financial data.

4.9 Lack of Large-Scale Validation

Many research prototypes require further testing with larger groups and longer periods.

Real-world evaluation is important to determine whether the system actually improves financial behaviour.

5. RESEARCH GAP AND UNRESOLVED ISSUES

The literature analysis identifies several important research gaps.

5.1 Integration Gap

Existing research often focuses on individual features such as:

- Expense tracking.
- Expense classification.
- Budget forecasting.
- Spending prediction.
- AI recommendations.

There is a need for a single integrated system combining these functions.

Proposed Solution

The proposed system combines:

Income Management + Expense Management + Budget Planning + Spending Analysis + AI Recommendations

5.2 Personalization Gap

Many existing systems use generic recommendations.

The proposed system can personalize recommendations based on:

- Income.
- Expenses.
- Previous spending.
- Budget.
- Savings goals.

5.3 Explainability Gap

AI recommendations should not simply provide an answer.

The system should explain the reason.

For example:

Recommendation: Reduce entertainment spending.

Explanation: Entertainment expenses increased from ₹3,000 to ₹5,000 compared with the previous month.

This makes the AI recommendation understandable.

5.4 New User Data Gap

Users with limited financial history may not receive accurate predictions.

Proposed Improvement

Use basic financial rules initially and gradually introduce machine-learning predictions as more transaction history becomes available.

5.5 Adaptive Budgeting Gap

Most budgeting systems use fixed limits.

A better approach is to dynamically recommend budget limits based on changing spending behaviour.

5.6 Security and Privacy Gap

Financial AI systems must provide intelligent recommendations while protecting sensitive user information.

This creates a need for secure data storage and access control.

6. POSSIBLE IMPROVEMENTS AND FUTURE RESEARCH DIRECTIONS

6.1 Advanced Machine Learning

Future versions can evaluate:

- Random Forest
- Decision Tree
- XGBoost
- LightGBM
- LSTM
- Transformer models

Different models can be compared using:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- MAE.
- RMSE.

6.2 Explainable AI

Explainable AI can show why a recommendation was generated.

This can improve:

- Transparency.
- User trust.
- Understanding.
- Financial awareness.

6.3 Automatic Receipt Processing

OCR can be added to extract information from receipts.

The system could automatically identify:

- Merchant.
- Date.
- Amount.
- Items.
- Category.

6.4 Conversational AI

A future version can provide a chatbot.

Example:

User: "How much did I spend on food this month?"

AI: "You spent ₹6,500 on food this month, which is ₹1,000 higher than last month."

6.5 Adaptive Budget Recommendations

The system can automatically adjust recommended budgets according to spending patterns.

For example:

If food expenses continuously exceed the budget, the system can:

1. Identify the trend.
2. Notify the user.
3. Suggest a realistic budget.
4. Recommend possible savings.

6.6 Financial Goal Planning

The system can allow users to create goals such as:

- Emergency fund.
- Travel.
- Education.
- Vehicle.
- Home.
- Savings.

AI can calculate suggested monthly savings.

6.7 Privacy-Preserving AI

Future research can investigate:

- Encryption.
- Federated Learning.
- Privacy-preserving machine learning.
- Secure authentication.
- Access control.

6.8 Long-Term Evaluation

Future research should measure whether AI recommendations actually improve:

- Saving behaviour.
- Budget adherence.
- Financial awareness.
- Spending discipline.

7. COMPARATIVE LITERATURE SURVEY TABLE

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Agarwal et al. (2024)	Lack of personalized finance assistance	AI, NLP	ML, React, MongoDB, Firebase	Personalized financial assistance	Limited large-scale validation	Need integrated real-world system
Kumar et al. (2024)	Difficulty predicting future spending	LSTM	Historical financial data	Predictive financial analysis	Requires historical data	Need adaptive recommendations
Bhavana et al. (2025)	Reactive financial management	ML, forecasting, XAI	Financial data	Personalized budget planning	Limited pilot evaluation	Need larger validation

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Amrendra (2025)	Manual expense tracking	ML classification	Expense data	Automated tracking and prediction	Limited generalization	Need stronger personalization
Zothansiam et al. (2025)	Lack of budget recommendations	Heuristic trend analysis	+ Transaction data	Budget and savings suggestions	Lightweight prediction	Need advanced adaptive AI
Garugu et al. (2026)	Manual financial processing	ML forecasting	+ Transaction data	Budget forecasting	Data-quality dependency	Need integrated AI system
Saranya et al. (2026)	Manual expense entry	AI + ML + NLP + OCR	+ Transaction/receipt data	Automated tracking and prediction	OCR limitations	Need robust multimodal processing
Chahar et al. (2026)	Static budgeting	ML + NLP	Financial/user data	Dynamic financial insights	Requires further validation	Need long-term evaluation

8. CITATIONS AND REFERENCES

The following references are presented using a consistent **IEEE-style format**.

References

- [1] V. Agarwal, R. Ray, and N. Varghese, “An AI-Powered Personal Finance Assistant: Enhancing Financial Literacy and Management,” *Research Cell: An International Journal of Engineering Sciences*, 2024.
- [2] J. Kumar, A. Vishwakarma, N. Sainy, A. Kumar, and R. Aggarwal, “Personal Finance Manager with Predictive Analytics Using LSTM for Enhanced Financial Decision Making,” *International Journal of Research – Granthaalayah*, 2024.
- [3] B. B. R. Bhavana, D. Pavan, and G. D. T. H. Guru Darshan, “AI-Based Wealth Advisory System using Machine Learning and Predictive Analytics for Personalized Budget Planning,” *International Journal of Advanced Research in Computer Science & Technology*, 2025.
- [4] K. Amrendra, “Smart Pocket: A Machine Learning-Based Expense Tracker and Spending Predictor,” *International Journal of Research and Innovation in Applied Science*, 2025.
- [5] V. Zothansiam, Lalrindika, and A. Dhuri, “Personal Finance Tracker with Budget Recommendation AI,” *International Journal of Innovative Research in Computer and Communication Eng*

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25